

# Autonomous Synthesis of Cyberpunk Animated Developer Profiles: Multi-Layer SVG SMIL Vectorization, Dithering & CI/CD Pipeline

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**Abstract**

*This white paper presents the architecture, graphics algorithms, and autonomous prompt specification for generating production-grade, interactive Cyberpunk Animated GitHub Profiles. By synthesizing raw developer artifacts (Resume PDFs, portrait photographs, and public Git repositories), the system autonomously compiles theme-adaptive, self-contained SVG SMIL graphics. Key technical innovations include an Atkinson error-diffusion pointillism converter, a dual-image 12-second holographic laser-sweep morphing mechanism, a 53-week active contribution snake game simulator, and automated cron-driven GitHub Actions workflows. The entire system is packaged into a single master prompt enabling deterministic 100% replication across any AI agent environment.*

## 1. System Inputs & Operating Principles

The generator operates on **three zero-dependency inputs** provided by the end developer, as outlined in Table 1:

| Input Artifact              | Extracted Properties & Schema                                                           | Downstream Utilization                                 |
|-----------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------|
| 1. Resume / CV (PDF/Text)   | Full Name, Professional Title, Education, ToolChain, Core Stack, Repositories, Contact. | Populates animated terminal telemetry & project cards. |
| 2. Portrait Photo (JPG/PNG) | Facial geometry, skin tone curves, shoulder contours, and high-frequency edge data.     | Dual-image base64 layer & Atkinson pointillism matrix. |
| 3. GitHub Handle            | Case-sensitive account string (e.g. Shaik6184).                                         | Special profile repo target & dynamic badge API links. |

## 2. The Universal Master AI Prompt Specification

The following self-contained prompt constitutes the complete executable instruction set. Feeding this prompt with the three input artifacts into any advanced AI agent reproduces the entire project deterministically:

**[UNIVERSAL AI MASTER PROMPT: CYBERPUNK ANIMATED GITHUB PROFILE BUILDER]****SYSTEM ROLE:**

You are an expert Principal Frontend Systems Architect, SVG Graphics Specialist, and Full-Stack Engineer. Your goal is to build an interactive, production-ready, theme-adaptive **Cyberpunk Animated GitHub Profile README** tailored 100% to the user's uploaded resume, portrait photo, and GitHub handle.

**INPUTS PROVIDED BY USER:**

1. Resume (Extract: Full Name, Role Title, Tech Stack, Education, Contact, and Repositories).
2. Portrait Photo.
3. GitHub Username (<USERNAME>).

**TECHNICAL ARCHITECTURE & DELIVERABLES TO BUILD:****1. THEME-AWARE DUAL-IMAGE MORPHING HERO BANNER (dark.svg & light.svg):**

- **Canvas Dimensions:** 1180x610 viewBox with 18px rounded macOS terminal frame, window controls (red, yellow, green), and animated glowing border (<linearGradient id="accent">).
- **Left Visual Box ('VISUAL.MAP', 400x492):** Implement an automated 12-second seamless looping morphing transition between:
  - **Layer 1 (Real Photo):** User's portrait cropped and aligned to head/chest, embedded as base64 with clip-path.
  - **Layer 2 (Cyberpunk Dither Matrix):** High-precision Atkinson error-diffusion pointillism portrait with 25,000+ vector dots.
  - **Transition Sweep:** Animated holographic laser scanline (<line>) sweeping top-to-bottom at 5s-6.5s and returning at 10.5s-12s.
  - **Traveling Vector Particles:** 480 SMIL particle vectors (<use href="#tvdark"> / <use href="#tvlight">) looping continuously across facial contour coordinates.
- **Right Terminal Bio ('SYSTEM.INFO'):** Render animated bio rows with staggered slide-in animations (<animateTransform>):  
Subject, Role, Origin, Education, Status, ToolChain, Core.Lang, Core.Frontend, Core.Backend, Core.Database, Core.Infra, and Contact (Mail, Phone, LinkedIn, GitHub, Portfolio).
- **GitHub Rendering Constraint:** All animations must use pure SVG SMIL (<animate>, <animateTransform>, <set>) with no external scripts so GitHub's sanitizer renders it natively.

**2. 53-WEEK ANIMATED CONTRIBUTION SNAKE GAME (snake-dark.svg & snake-light.svg):**

- Construct a full 53-week × 7-day contribution grid with dense, realistic commit activity across all months.
- Apply theme palettes: Dark (#0A101F bg, indigo/purple/violet/cyan/emerald dots) and Light (#F8FAFC bg, cyan/purple/emerald dots).
- Animate a 5-segment glowing snake actively patrolling and eating commits across the grid with a synchronized bottom progress bar.

**3. PROJECT SHOWCASE TERMINAL PANEL (projects.svg & projects-light.svg):**

- Python generator (.github/scripts/generate\_projects.py) rendering 6 curated repositories into card panels with custom badges, stars, and language byte percentages.

**4. GITHUB PROFILE README.md & ACTIONS WORKFLOWS:**

- Assemble README.md with theme-aware <picture> elements, streak stats, top language split, snake game, project cards, and social badges.
- Configure .github/workflows/projects.yml and snake.yml for automated daily cron updates.
- Initialize Git, commit all assets, and push to main & output branches of github.com/<USERNAME>/<USERNAME>.

### 3. Mathematical Formulation of Graphics Engines

#### 3.1 Atkinson Error-Diffusion Halftone Algorithm

Standard Floyd-Steinberg dithering diffuses 100% of quantization errors, which can produce muddy noise in low-contrast facial regions. In contrast, the **Atkinson Dithering Matrix** diffuses only 75% of error across a 6-neighbor kernel, discarding the remaining 25% to generate crisp high-key highlights and clean pointillist contours:

**Atkinson Discrete Convolution Kernel:**

- $e_{x+1, y} \leftarrow e_{x+1, y} + 1/8 \times \Delta v$
- $e_{x-1, y+1} \leftarrow e_{x-1, y+1} + 1/8 \times \Delta v$
- $e_{x+1, y+1} \leftarrow e_{x+1, y+1} + 1/8 \times \Delta v$
- $e_{x+2, y} \leftarrow e_{x+2, y} + 1/8 \times \Delta v$
- $e_{x, y+1} \leftarrow e_{x, y+1} + 1/8 \times \Delta v$
- $e_{x, y+2} \leftarrow e_{x, y+2} + 1/8 \times \Delta v$

where  $\Delta v = v_{old} - v_{quantized}$  and  $v_{quantized} \in \{0, 255\}$ .

#### 3.2 Dual-Image Holographic Morph Timing Function

The morphing engine alternates between raster photo fidelity and cyber matrix abstraction across a deterministic 12-second periodicity ( $T = 12.0s$ ):

| Interval      | State & Active Layer         | Laser Position (Y) | Visual Description                                   |
|---------------|------------------------------|--------------------|------------------------------------------------------|
| 0.0s – 5.0s   | Real Photo (Opacity: 1.0)    | 84 px (Hidden)     | High definition portrait with subtle radar sweep.    |
| 5.0s – 6.5s   | Transition 1: Photo → Dither | 84 → 576 px        | Laser sweeps downwards; photo dissolves into dither. |
| 6.5s – 10.5s  | Cyber Matrix (Opacity: 1.0)  | 576 px (Hidden)    | Pointillism matrix + 480 continuous particle loops.  |
| 10.5s – 12.0s | Transition 2: Dither → Photo | 576 → 84 px        | Reverse sweep; dither dissolves, restoring photo.    |

### 4. Step-by-Step GitHub Setup & Deployment Pipeline

| Phase                   | Procedure / Terminal Command                                                                                                                                    |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Special Repo Setup   | Create a new public repository on GitHub named exactly <code>https://github.com/&lt;USERNAME&gt;/&lt;USERNAME&gt;</code> .                                      |
| 2. SVG Compilation      | Run local Python compilation scripts to generate <code>dark.svg</code> , <code>light.svg</code> , <code>snake-dark.svg</code> , and <code>projects.svg</code> . |
| 3. Push to GitHub       | <code>git remote add origin https://github.com/&lt;USER&gt;/&lt;USER&gt;.git &amp;&amp; git push -u origin main --force</code>                                  |
| 4. Actions Permissions  | Navigate to <b>Repo Settings</b> → <b>Actions</b> → <b>General</b> → <b>Workflow permissions</b> , enable <b>'Read and write permissions'</b> , and click Save. |
| 5. Profile Verification | Visit <code>https://github.com/&lt;USERNAME&gt;</code> . GitHub automatically displays the dark/light SVG banner, streak stats, snake game, and project panel.  |

**Document Certification:** This technical white paper specifies the complete self-contained pipeline for automated cyberpunk GitHub profile generation. Upload this PDF to Google Drive or share with engineering teams for immediate replication.